🚀 ClinikFlow: Step-by-Step Phased Implementation Checklist

Product Name: ClinikFlow (India Specialist Edition)
File: `docs/05_PHASED_IMPLEMENTATION_CHECKLIST.md`
Target: Standalone Single-Doctor Specialist Outpatient Clinics in India
Status: Living Engineering Checklist & Progress Tracker

🗺️ Master Phase Overview

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Phase 1: High-Speed Front-Desk (The "Compounder Desk") ──────→ Weeks 1 - 2
Phase 2: Doctor Mobile Cabin Controller & TV Display ──────→ Weeks 3 - 4
Phase 3: WhatsApp Automation & The "Report Check" Queue ──────→ Weeks 5 - 6
Phase 4: Specialist Clinical EMR Lite & Rapid Rx Engine ──────→ Weeks 7 - 8
Phase 5: Dynamic UPI Payments & Daily OPD Ledger ──────→ Weeks 9 - 10
Phase 6: Offline Resilience, Pilot Deployment & Field UAT ──→ Weeks 11 - 12
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⚡ Phase 1: High-Speed Indian Front-Desk (The "Compounder Desk") ✅ COMPLETED

Objective: Enable the compounder to register walk-ins and issue a token in < 12 seconds.

- ✓ **1.1. Optimize Registration UI for Speed (`js/views/registerView.js`)**
  - ✓ Add auto-focus to the Phone Number input upon view mount.
  - ✓ Implement debounced lookup on 10-digit Indian mobile number: if patient exists, auto-populate Name, Age, Gender, and UHID instantly.
  - ✓ Enable full keyboard-driven navigation (`Alt+N` for new token, `Enter` to confirm submission, `Esc` to cancel).
- ✓ **1.2. Priority Triage & Category Tagging**

    - ✓ 1-tap segment selector: `Walk-in (Normal)`, `Scheduled Appt`, `Emergency`, `Senior Citizen / Infant`.
    - ✓ Support chronological daily token numbering (Token #1, #2, #3 resets at midnight via `getNextTokenNumber`).
- ✓ **1.3. State & Sync Hardening (`js/core/state.js`)**
    - ✓ Wire multi-tab `BroadcastChannel('clinik_local_sync')` so new tokens broadcast immediately without full-page reloads.
    - ✓ Store daily tokens with payment status (`PAID_CASH`, `PAID_UPI`, `PENDING`) and fee amount.
- ✓ **1.4. Indian Clinic Workflow: "Hold for Lab / Report Review"**
    - ✓ `holdForLab(patientId)` status transition.
    - ✓ `markReportsArrived(patientId)` re-entry priority queue.
    - ✓ TV Kiosk update with bilingual text, wake lock, and token number display.

📱 Phase 2: Doctor Mobile Cabin Controller & Smart TV View

Objective: Empower the doctor to run the OPD from an iPad/phone and broadcast to the waiting room.

- **2.1. Dedicated Doctor Mobile View (`js/views/doctorPadView.js`)**
    - Build a sticky top queue ribbon optimized for mobile thumb reach:
        - Current Patient card with timer.
        - Large 54px action buttons: `[ Call Next ]`, `[ Hold for Lab ]`, `[ Complete ]`.
    - Add slide-over drawer to inspect the next 3 waiting patients.
- **2.2. Waiting Room Smart TV Enhancements (`tv-display.html`)**
    - Implement a 25-foot viewing mode with large typography and dynamic background color pulses when a new token is called.
    - Integrate screen wake lock API (`navigator.wakeLock.request('screen')`) so TV screens never sleep or go dim during clinic hours.

- **2.3. Dual-Language Audio Calling (`js/core/audio.js`)**
  - Enhance Web Audio chime ($E_5 \rightarrow C_5$ E5 → C5).
  - Add speech synthesis (Web Speech API) for voice announcements:
    - English: "*Token number 15, please proceed to Doctor's Cabin.*"
    - Hindi: "*टोकन नंबर 15, कृपया डॉक्टर केबिन में आएं।*"

💬 Phase 3: WhatsApp Dispatch & The "Report Check" Queue

Objective: Eliminate waiting room chaos with live patient tracking and seamless diagnostic report returns.

- **3.1. Zero-App Patient Tracking Web Page (`patient-queue.html`)**
  - Build a lightweight (<100KB) mobile tracker accessible via link (`clinic.in/q/{token_id}`).
  - Live counter: "Tokens ahead of you: 3", "Estimated wait time: ~18 mins".
  - Dynamic color shift: Blue (waiting) → → Green (top 2) → → Gold (inside cabin).
  - Add "Running 15 mins late" button allowing patient to slide back by 2 slots.
- **3.2. Automated WhatsApp Cloud API Webhook**
  - Dispatch template message on token generation with tracking URL and live position.
  - Send alert when patient reaches position #2: "*Please proceed to the clinic waiting area.*"
- **3.3. The "Report Check / Re-Entry" Workflow**
  - Doctor taps `[Hold for Lab]` → → status moves to `AWAITING_REPORTS`.
  - When patient returns with lab/X-ray results, compounder taps `[Mark Reports Arrived]`.
  - System automatically interleaves the patient into the next priority slot without re-billing.

🩺 Phase 4: Specialist Clinical EMR Lite & Rapid Rx Engine

Objective: Enable the doctor to record vitals and generate Indian-format prescriptions in < 60 seconds.

- **4.1. Universal Vitals Strip**
  - Quick numpad entry for BP (`Systolic / Diastolic`), Pulse, Temp, Weight, Height (auto-computes BMI), and Blood Sugar.
- **4.2. Specialist-Specific Clinical Micro-Modules**
  - **Pediatrics:** IAP immunization milestone checklist + Age-for-weight percentile badges.

    - **Dermatology:** Interactive tap-to-select body map + lesion morphology chips (`Erythema`, `Plaque`, `Macule`, `Pruritus 1-10`).
    - **Orthopedics:** Joint locator (`Knee`, `Spine`, `Shoulder`) + range-of-motion toggles + common X-ray order buttons.
    - **Diabetology:** Comorbidity pills (`Type 2 DM`, `Hypertension`, `Dyslipidemia`) + HbA1c history tracker.
- **4.3. Rapid Indian Prescription (Rx) Engine**
    - Autocomplete database of top Indian brand names & compositions (e.g. *Augmentin 625*, *Pan-D*, *Calpol 250*).
    - One-tap dosage chips: `1-0-1`, `1-0-0`, `0-0-1`, `1-1-1`, `SOS`.
    - One-tap meal timing: `After Food (खाने के बाद)`, `Before Food (खाली पेट)`.
    - Auto-compile digital PDF with clinic letterhead and doctor registration number for 1-tap WhatsApp sharing.

💳 Phase 5: Dynamic UPI Payments & Financial Settlement

Objective: Reconcile cash and digital payments seamlessly at the front desk.

- **5.1. Dynamic UPI QR Generator**
    - Generate dynamic BharatQR code with embedded clinic VPA (`doctor@upi`) and exact consultation fee amount.
    - Patient scans with Google Pay / PhonePe / Paytm; receptionist verifies with 1-click confirmation.
- **5.2. ESC/POS Thermal Slip Printing**
    - Web Bluetooth / Web USB driver for 2-inch and 3-inch thermal printers.
    - Print token slips with large bold token number and queue tracking QR code.
- **5.3. Daily Cash Drawer & OPD Settlement Ledger**
    - End-of-day summary: Total Patients, Cash vs. UPI split, average consultation duration, and abandoned tokens.
    - 1-click export of settlement report to the doctor's WhatsApp.

🛡️ Phase 6: Offline Resilience, UAT & Live Pilot

Objective: Ensure 100% clinic continuity during internet outages and conduct pilot testing.

- **6.1. Service Worker & IndexedDB Offline Queue**
  - Cache all static assets (HTML/CSS/JS) for full offline boot.
  - Queue mutations locally in `IndexedDB` when offline; auto-sync to backend when connection is restored.
- **6.2. 3-Day Shadow Pilot**
  - Deploy at a pilot clinic running alongside the paper register for 3 days.
  - Verify that front-desk registration takes < 12 seconds and doctor Rx writing takes < 60 seconds.
- **6.3. Cutover to 100% Digital Clinic**
  - Remove paper register and rely entirely on ClinikFlow.